

The Trial

AP Statistics · trial design

One long floor, where walking north is distance from the patient.

45	24	3.2	6/11	3	26 · 18 · 1
STOPS	CHOICE (53%)	EFFECTIVE FORMATS	EQUATIONS COMPUTED	SELECT-ONLY CONCEPTS	SELECT · CONSTRUCT · OPERATE

READING Where this campaign stands

Twenty-four CHOICE stops out of 45 and 5 of 11 equations computed. Three gaps sit on instruments and may clear in Phase 0; three do not, and they are the three that decide whether a trial can answer its question — alpha spending, the events formula, and conditional power. The place is the hardest constraint in the set: one long floor, no vehicles, and the walk down it is distance from the patient. That rules out the driven formats and leaves the ones that run on data, which is fortunate, because this is AP Statistics.

NOW The mix as it ships

CHOICE 24	BALLPARK 4	PROTOCOL 4	SEQUENCE 3	TRIAGE 2	VALUE 2	CONTROL 1	ALLOCATE 1	ATTEST 1	TRACE 1	DELEGATE 1
DEGENERACY 1										

CURRICULUM What the syllabus is not getting

EQUATION THE COURSE OWES	WHAT IT CARRIES	MENTIONED AT	STATUS
$RR = p_{\text{treatment}} / p_{\text{control}}$	relative risk, and why it hides how big the effect is	29 · BALLPARK 43 · DEGENERACY	verify first
$CI \approx \text{estimate} \pm 1.96 \times SE$	a confidence interval, and what its width says	29 · BALLPARK 36 · VALUE 44 · CHOICE	verify first
$\sum \alpha_{\text{spent}} \leq \alpha$	alpha spending — a budget for error that does not refill	10 · CHOICE 40 · PROTOCOL 41 · CHOICE 44 · CHOICE	real gap
$\text{events} \approx 4 (z_{\{\alpha/2\}} + z_{\beta})^2 / (\ln HR)^2$	how many events a trial has to see to answer its question	16 · BALLPARK 20 · CHOICE 40 · PROTOCOL 41 · CHOICE	real gap
conditional power = P(significant at the end data so far)	what a futility forecast is, and what it is not	40 · PROTOCOL 41 · CHOICE	real gap

2 of 5 sit on an instrument stop, whose board arithmetic the coverage check cannot yet read — expect Phase 0 to clear those. 3 are unaffected by it.

TIER Concepts the player only ever picks from a list

#	MECHANISM CONCEPT	STOPS
8	The primary endpoint, named in advance, and everything else	2, 44
22	Intention to treat against per-protocol	20
29	Unblinding an individual for their own care	12

Reported, never failed. Select-tier can be exactly right; these are the candidates, not the defects.

SWEEP STRONG

stop 29 or 34 · STAT

Drag the sample size and watch the interval narrow — and it narrows as the square root, which is the one thing about precision every student gets wrong. "How big, and how wide" is currently a BALLPARK; the curve is the answer and it is plotted only where you look.

SERVES $SE = \sqrt{p(1-p)/n}$ and $CI \approx \text{estimate} \pm 1.96 \times SE$ — two gaps in one panel

NEEDS the diminishing return visible in the shape, not stated in the hint

TALLY STRONG

stop 16 · STAT

Events accumulate and the analysis becomes possible. "How much of the trial has happened" is a BALLPARK about exactly this, and TALLY's own subject — when is there enough to report — **is** interim analysis.

SERVES the events formula and alpha spending

NEEDS an early look that flatters the result

SPOT STRONG

stop 10 or 40 · STAT / SAFE

A stopping rule replaced mid-trial with nothing announcing it. This is the sharpest statistical version of the format anywhere: an unannounced change to a stopping rule is how alpha gets spent without anybody deciding to spend it.

SERVES $\sum \alpha_{\text{spent}} \leq \alpha$ — a budget for error that does not refill

RISK score only the discriminating items

HOLDOUT STRONG

stop 34 · STAT

"The subgroup that crossed" is a CHOICE, and subgroup analysis is the canonical overfitting story in clinical research. Fit the subgroup rule on one half, freeze it, score it on the half it never saw, and the flattering answer costs the stop.

SERVES multiplicity and subgroup discipline

NEEDS a spike in the fitting curve that beats the honest answer

BELT MEDIUM

SITE · monitoring

Case report forms sorted against one criterion — protocol deviation or not — at the speed a monitoring visit runs. Real trial work, and one honest bit. Medium because the criterion in practice needs more than one clause.

RISK a definition needing two clauses is not a BELT

TRIAL DEFER

the long floor

The corridor is walkable and its length is the theme, so a driven order is not absurd. Deferred because a coordinating centre has nothing to drive and the gates would be doors.

HOLD REJECT

No physical quantity to hold. Reject on geometry.

LOB REJECT

Reject.

SLATES Three appetites — the campaign is zero-sum

No conversion adds a stop. Each changes what an existing stop **does** while leaving what it **teaches** alone. Snapshot before, diff after: `curriculumDelivery.mjs the_trial --snapshot before.json`, then `--against before.json`.

Minimum honest 3 STOPS · ONE SITTING

29	d10	STAT	BALLPARK → SWEEP	How big, and how wide how big and how wide — the square root, as a shape
16	d6	STAT	BALLPARK → TALLY	How much of the trial has happened events accumulated until the interim can be read
34	d12	STAT	CHOICE → HOLDOUT	The subgroup that crossed the subgroup that crossed, scored on data it never saw

CHOICE 24 → 23 (51%) · moves 26·18·1 → 25·16·4 · effective formats 3.2 → 3.5 · 3 new operate stops

Balanced 6 STOPS · THE RECOMMENDATION

29	d10	STAT	BALLPARK → SWEEP	How big, and how wide
16	d6	STAT	BALLPARK → TALLY	How much of the trial has happened
34	d12	STAT	CHOICE → HOLDOUT	The subgroup that crossed
10	d4	STAT	CHOICE → SPOT	What an extra look spends what an extra look spends, under a rule that changes
20	d7	STAT	CHOICE → ATTEST	Everyone as assigned everyone as assigned — the record is not the condition
2	d1	ENDP	CHOICE → TRIGGER	One outcome, named in advance one outcome named in advance, written as a rule before the data moves

CHOICE 24 → 20 (44%) · moves 26·18·1 → 22·18·5 · effective formats 3.2 → 4.5 · 4 new operate stops, 1 of them arcade

Ambitious 9 STOPS · ADDS WORLD WORK

29	d10	STAT	BALLPARK → SWEEP	How big, and how wide
16	d6	STAT	BALLPARK → TALLY	How much of the trial has happened
34	d12	STAT	CHOICE → HOLDOUT	The subgroup that crossed
10	d4	STAT	CHOICE → SPOT	What an extra look spends
20	d7	STAT	CHOICE → ATTEST	Everyone as assigned
2	d1	ENDP	CHOICE → TRIGGER	One outcome, named in advance
41	d14	STAT	CHOICE → CLOUD	What the rest of the trial would probably do what the rest of the trial would probably do, as a band
44	d15	ENDP	CHOICE → BALANCE	What can be claimed what can be claimed — closed as a ledger
12	d4	RAND	CHOICE → BELT	When the blind is opened for one person deviation or not, at monitoring-visit speed

CHOICE 24 → 17 (38%) · moves 26·18·1 → 19·20·6 · effective formats 3.2 → 5.9 · 5 new operate stops, 2 of them arcade

THE ORDER OF WORK

1. Phase 0 — teach the coverage check to read an instrument's own board, and re-baseline curriculum-debt.json. Rewriting a stop that already teaches the thing is the exact failure the check exists to prevent. 2. Snapshot. 3. Edit the book, never the generated content — tools/import-book.mjs books/the_trial.yml the_trial --verify. 4. npm run traps: every instrument's trap must fire. 5. npm run check the_trial and npm run drive the_trial, right answer and wrong — ask what a player who understood nothing would score before believing any pass mark. 6. --against the snapshot. A changed format is reported; a changed objective fails.

Counts computed from `themes/the_trial/content/` via `engine/dev/curriculumDelivery.mjs`; concept and equation lists authored in `tools/syllabus.js`. Format verdicts and slates are editorial. Run it with `THEME=the_trial npm run dev`.